



MEDICAL REFERENCE SERIES

EMERGENCY MEDICINE

PRACTICAL EMERGENCY CARE & QUICK
REFERENCE GUIDE

*Rapid Assessment • Emergency Protocols • Critical Care • Drug Doses • Quick
Reference*

FOR

Medical Students • Nursing Students • Interns • Paramedical Staff •
Emergency-Care Learners

Medical Reference Series | 2026

Important Medical Disclaimer

This guide is designed for educational and clinical-reference learning purposes.

Emergency treatment depends on the individual patient's age, weight, medical history, allergies, pregnancy status, current medications, investigations, available resources, and institutional protocols.

Drug names, doses, procedures, and treatment pathways included here should not be interpreted as individualized medical advice, or as permission for an untrained person to administer medication or perform a medical procedure.

Medication doses should be checked against current institutional protocols, product information, and applicable professional guidelines before clinical use.

⚠ WARNING

- In an actual emergency, activate emergency medical services or seek immediate treatment from a qualified healthcare professional.

How to Use This Guide

Emergency medicine requires a different mindset from routine clinical care. In many emergencies, the clinician cannot wait for a final diagnosis before beginning stabilization.

The Core Sequence

RECOGNIZE



PRIORITIZE



STABILIZE



INVESTIGATE



TREAT



REASSESS

How Each Chapter Is Organized

Every chapter in this guide follows the same six-part structure, so you always know where to find what you need:

Section	What it answers
Recognition	What does the emergency look like?
Assessment	What should be checked immediately?
Immediate Management	What should happen first?
Drug Reference	Common medicines and doses from the reference material.
Red Flags	What suggests severe disease?
Quick Recall	Important revision points.

Table of Contents

Emergency Medicine Essentials	1
1.1 What Is Emergency Medicine?	1
1.2 Three Immediate Objectives	1
1.3 Major Emergency Categories	2
1.4 The Time-Critical Principle	2
Chapter Summary	2
Key Takeaways	3
Primary Survey — ABCDE	4
2.1 A — Airway	4
2.2 B — Breathing	5
2.3 C — Circulation	6
2.4 D — Disability	6
2.5 E — Exposure	7
2.6 Reassessment	7
Chapter Summary	7
Key Takeaways	7
Cardiac Arrest & CPR	9
3.1 Recognition	9
3.2 Basic Sequence	9
3.3 High-Quality Adult Chest Compressions	10
3.4 Cardiac Arrest Rhythms	10
3.5 Cardiac Arrest Drugs	10
3.6 Reversible Causes — Hs & Ts	11
3.7 Return of Spontaneous Circulation (ROSC)	11
Chapter Summary	11
Key Takeaways	12
Shock & Circulatory Failure	13
4.1 Recognizing Shock	13
4.2 Major Categories	13

4.3 Hypovolemic Shock.....	14
4.4 Cardiogenic Shock	14
4.5 Septic Shock	15
4.6 Anaphylactic Shock.....	15
Chapter Summary.....	15
Key Takeaways	15
Respiratory Emergencies	17
5.1 Acute Asthma.....	17
5.2 Acute COPD Exacerbation	18
5.3 Pulmonary Edema.....	18
5.4 Tension Pneumothorax.....	19
5.5 Aspiration	19
5.6 Respiratory Failure.....	19
Chapter Summary.....	19
Key Takeaways	19
Cardiac Emergencies	21
6.1 Acute Coronary Syndrome.....	21
6.2 Tachyarrhythmias.....	21
6.3 Bradyarrhythmia.....	22
6.4 Acute Heart Failure	22
6.5 Hypertensive Emergency	22
6.6 Cardiac Tamponade.....	23
Chapter Summary.....	23
Key Takeaways	23
Neurological Emergencies.....	24
7.1 Seizures.....	24
7.2 Status Epilepticus	25
7.3 Stroke.....	25
7.4 Altered Sensorium / Coma.....	26
7.5 Raised Intracranial Pressure	26
7.6 Meningitis / Encephalitis	26

Chapter Summary	26
Key Takeaways	27
Poisoning & Overdose	28
8.1 Organophosphate Poisoning	28
8.2 Drug Overdose	29
8.3 Alcohol Poisoning	29
8.4 Corrosive Poisoning	29
8.5 Snakebite	30
Chapter Summary	30
Key Takeaways	30
Trauma Management	32
9.1 Head Injury	32
9.2 Chest Trauma	33
9.3 Abdominal Trauma	33
9.4 Musculoskeletal Trauma	33
9.5 Polytrauma / Major Hemorrhage	34
Chapter Summary	34
Key Takeaways	34
Pediatric Emergencies	36
10.1 Pediatric Assessment Triangle	36
10.2 Pediatric Respiratory Distress	36
10.3 Febrile Seizure	37
10.4 Pediatric Dehydration / Shock	37
10.5 Pediatric Sepsis	37
10.6 Pediatric Hypoglycemia	37
10.7 Pediatric Poisoning	38
Chapter Summary	38
Key Takeaways	38
Obstetric Emergencies	39
11.1 Postpartum Hemorrhage (PPH)	39
11.2 Eclampsia	40

11.3 Antepartum Hemorrhage	40
11.4 Ruptured Ectopic Pregnancy	41
11.5 Obstructed Labor	41
11.6 Sepsis in Pregnancy	41
Chapter Summary.....	42
Key Takeaways	42
Emergency Drug Quick Reference	43
12.1 Resuscitation	43
12.2 Respiratory	44
12.3 Cardiac.....	44
12.4 Neurological	44
12.5 Shock / Trauma.....	44
12.6 Antidotes	44
12.7 Obstetric	45
12.8 Metabolic.....	45
12.9 Analgesia	45
Chapter Summary.....	46
Key Takeaways	46
Emergency Investigations	47
13.1 Bedside Tests	47
13.2 Laboratory Tests	48
13.3 Imaging.....	48
13.4 Cardiac Investigations	48
13.5 Infection Workup	49
13.6 Time-Critical Quick View	49
Chapter Summary.....	49
Key Takeaways	49
Triage.....	51
14.1 The Four Triage Categories	51
14.2 START Triage.....	52
14.3 Pediatric Triage.....	52

14.4 Re-Triage.....	52
Chapter Summary.....	52
Key Takeaways	52
Medico-Legal Essentials.....	54
15.1 Documentation	54
15.2 Consent	55
15.3 Police Intimation	55
15.4 Evidence Preservation	55
Chapter Summary.....	56
Key Takeaways	56
Emergency Medicine Career Pathways	57
16.1 After MBBS.....	57
16.2 Postgraduate Training.....	58
Chapter Summary.....	58
Key Takeaways	58
Rapid Emergency Reference	60
The First-Minutes Checklist.....	60
Cardiac Arrest Quick Card	61
Shock — Quick Differentiation	62
Stroke Quick Card.....	62
Seizure Quick Card	62
Poisoning Quick Card	63
Trauma Quick Card	63
Pediatric Quick Card	64
Obstetric Red Flags.....	64
Emergency Drug Snapshot.....	64
Final Takeaway.....	65

CHAPTER 1

Emergency Medicine Essentials



1.1 What Is Emergency Medicine?

Emergency medicine focuses on the rapid recognition, assessment, and stabilization of patients with sudden illness or injury. The immediate goal is often not to establish every detail of the diagnosis. The first question the clinician asks is:

“What can seriously harm or kill this patient in the next few minutes?”

The clinician then identifies and treats those threats while the diagnostic process continues in parallel.

1.2 Three Immediate Objectives

1. Preserve Life — identify conditions that threaten airway, breathing, circulation, or neurological function.
2. Stabilize the Patient — support essential physiological functions while determining the underlying cause.
3. Prevent Secondary Damage — prolonged hypoxia, shock, hemorrhage, and delayed intervention can produce additional organ injury.

1.3 Major Emergency Categories

Emergency presentations can broadly involve:

- Cardiac emergencies
- Respiratory emergencies
- Neurological emergencies
- Trauma
- Poisoning
- Pediatric emergencies
- Obstetric emergencies
- Metabolic emergencies
- Severe infections

Emergencies are further distinguished as medical, surgical, trauma, pediatric, and obstetric in nature — a classification that shapes how a department organizes its response.

1.4 The Time-Critical Principle

Emergency medicine is fundamentally time-sensitive. Rapid recognition does not mean rushing without structure — it means using a systematic approach so that the most dangerous abnormality receives attention first.

★ KEY POINT

- Treat immediate threats while assessment continues — stabilization and diagnosis happen together, not in sequence.

Chapter Summary

Emergency medicine prioritizes immediate threats to life over a complete diagnosis. Every emergency presentation is approached with three goals in mind — preserving life, stabilizing physiology, and

preventing further organ damage — using a structured, time-aware approach rather than an unstructured rush.

Key Takeaways

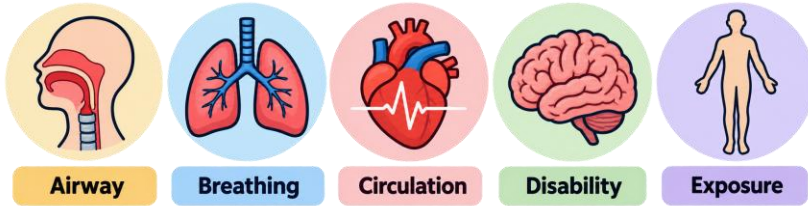
- Ask “what could kill this patient in the next few minutes?” before asking “what is the diagnosis?”
- Preserve life, stabilize, and prevent secondary damage — in that order of urgency.
- Emergencies span medical, surgical, trauma, pediatric, and obstetric categories.
- Speed without structure is dangerous; use a systematic sequence instead.

✓ QUICK RECALL

- What are the three immediate objectives of emergency care?
- Why can secondary organ damage occur even after the primary problem is treated?
- Name three major categories of emergency presentation.

CHAPTER 2

Primary Survey — ABCDE



ABCDE provides a structured approach to the critically ill patient, used as the primary survey for recognizing and treating immediate threats to life.

A — Airway**B — Breathing****C — Circulation****D — Disability****E — Exposure**

2.1 A — Airway

Ask: is the airway patent and protected? Possible warning signs include:

- Inability to speak normally
- Stridor or abnormal airway sounds
- Blood or vomit
- Foreign material

- Reduced consciousness
- Loss of airway protection

Initial Airway Measures

For appropriately trained personnel, airway management may include:

- Head tilt–chin lift — for a non-traumatic situation
- Jaw thrust — when trauma or cervical injury is suspected

Possible adjuncts include an oropharyngeal airway, a nasopharyngeal airway, or endotracheal intubation when indicated.

Drug	Dose	Use
Lignocaine spray	Topical	Before intubation
Midazolam	0.05–0.1 mg/kg IV	Sedation for airway procedures

2.2 B — Breathing

Assess respiratory rate, respiratory effort, chest movement, oxygen saturation, cyanosis, and air entry. Consider immediately dangerous respiratory problems such as severe asthma, pneumothorax, pulmonary edema, and respiratory failure.

Oxygen Delivery

Device	Flow Rate
Nasal cannula	2–5 L/min
Face mask	6–10 L/min
Non-rebreather mask	10–15 L/min

Common Respiratory Drugs

Drug	Dose
Salbutamol	2.5–5 mg nebulized
Ipratropium bromide	0.5 mg nebulized

Drug	Dose
Hydrocortisone	100 mg IV
Magnesium sulfate (severe cases)	2 g IV slowly

2.3 C — Circulation

Assess pulse rate and quality, blood pressure, peripheral perfusion, capillary refill, and external hemorrhage. Look for possible shock, major blood loss, severe dehydration, or cardiac dysfunction.

Immediate Measures

Establish vascular access when required — two large-bore 16–18G IV cannulas are recommended in relevant critically ill patients.

Drug / Fluid	Dose	Use
Normal saline / Ringer lactate	20 mL/kg IV	Initial fluid in shock
Adrenaline infusion	0.05–0.3 mcg/kg/min IV	Vasopressor support
Noradrenaline infusion	0.05–0.3 mcg/kg/min IV	Vasopressor support

Control significant external hemorrhage using direct pressure and pressure dressings.

2.4 D — Disability

Perform a rapid neurological assessment: GCS, pupils, blood glucose, movement/weakness, and seizure activity. Potential causes of altered neurological function include head injury, stroke, hypoglycemia, and drug overdose.

Drug	Dose	Use
Dextrose 25%	100 mL IV slowly	Hypoglycemia
Diazepam	10 mg IV slowly	Seizure

Drug	Dose	Use
Lorazepam	4 mg IV	Seizure

2.5 E — Exposure

Examine sufficiently to identify hidden bleeding, burns, fractures, rashes, and other injuries. Prevent hypothermia after examination — remove wet clothing when appropriate and keep the patient warm.

2.6 Reassessment

ABCDE is a cycle, not a one-time checklist.

Repeat ABCDE



Recheck vitals



Assess response



Escalate if necessary

Chapter Summary

The primary survey moves through Airway, Breathing, Circulation, Disability, and Exposure in strict order, treating each life-threat as it is found rather than waiting to finish the full examination. Reassessment after every intervention keeps the survey a continuous cycle rather than a single pass.

Key Takeaways

- Fix problems in the order they are found — airway before breathing, breathing before circulation.
- Oxygen delivery device and flow rate should match the severity of respiratory compromise.
- Two large-bore IV lines are the default in a critically ill patient.

- Blood glucose is a mandatory check in any patient with altered neurological status.
- ABCDE does not end after one pass — reassess after every intervention.

✓ **QUICK RECALL**

- What are the five components of the primary survey, in order?
- What oxygen flow rate matches a non-rebreather mask?
- What is the initial IV fluid dose used in a hypotensive, poorly perfused patient?

CHAPTER 3

Cardiac Arrest & CPR



3.1 Recognition

Suspect cardiac arrest when a patient is unresponsive, not breathing normally or only gasping, and without a detectable pulse on an appropriate rapid check. Cardiac arrest requires immediate resuscitation.

3.2 Basic Sequence

1. Ensure scene safety — protect the patient, rescuer, and surrounding team.
2. Check responsiveness — assess quickly.
3. Activate emergency response — call for help or the resuscitation team and obtain a defibrillator/AED where available.
4. Assess breathing and pulse — a pulse check should last no longer than approximately 10 seconds.

3.3 High-Quality Adult Chest Compressions

Parameter	Target
Rate	100–120 / minute
Depth	5–6 cm
Compression-to-ventilation ratio	30:2 (single rescuer)

Allow complete chest recoil between compressions and minimize interruptions.

3.4 Cardiac Arrest Rhythms

Shockable

- Ventricular fibrillation
- Pulseless ventricular tachycardia

Non-shockable

- Asystole
- Pulseless electrical activity

Defibrillation

★ KEY POINT

- Biphasic defibrillation: 200 J for VF / pulseless VT, followed by immediate resumption of CPR.

3.5 Cardiac Arrest Drugs

Drug	Dose	Indication
Adrenaline	1 mg IV, repeat every 3–5 min	Cardiac arrest rhythms
Amiodarone	300 mg IV bolus, then 150 mg IV	Refractory VF / VT
Sodium bicarbonate	50 mEq IV	Severe acidosis, hyperkalemia

Drug	Dose	Indication
Magnesium sulfate	2 g IV slowly	Torsades de pointes

3.6 Reversible Causes — Hs & Ts

Hs

- Hypoxia
- Hypovolemia
- Hydrogen ion / acidosis
- Hypo-/hyperkalemia
- Hypothermia

Ts

- Tension pneumothorax
- Cardiac tamponade
- Toxins
- Coronary/pulmonary thrombosis

3.7 Return of Spontaneous Circulation (ROSC)

ROSC may be suggested by a detectable pulse, measurable blood pressure, spontaneous breathing, or other signs of restored circulation. Continue post-arrest monitoring and identify the underlying cause.

Chapter Summary

Cardiac arrest management follows a fixed sequence: confirm the arrest, start high-quality compressions, identify the rhythm, defibrillate shockable rhythms, and give drugs on a timed cycle while actively searching for a reversible cause.

Key Takeaways

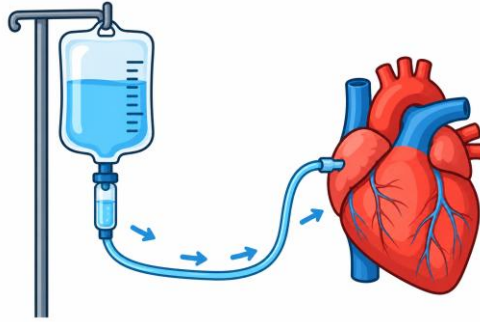
- Confirm unresponsiveness, abnormal/absent breathing, and absent pulse within about 10 seconds.
- Compress at 100–120/min, 5–6 cm deep, with a 30:2 ratio for a single rescuer.
- Shockable rhythms get 200 J biphasic defibrillation; non-shockable rhythms do not.
- Adrenaline 1 mg IV every 3–5 minutes applies to all arrest rhythms.
- Always search for the Hs and Ts — treating the reversible cause can be the difference between ROSC and continued arrest.

✓ QUICK RECALL

- What is the correct compression rate and depth for adult CPR?
- What energy is used for biphasic defibrillation of VF?
- List the five Hs and four Ts of reversible cardiac arrest causes.

CHAPTER 4

Shock & Circulatory Failure



Shock is inadequate tissue perfusion leading to insufficient oxygen delivery to organs. Importantly, shock can exist before obvious hypotension develops.

4.1 Recognizing Shock

Possible signs include:

- Tachycardia
- Hypotension
- Cold / clammy skin
- Reduced urine output
- Altered mental status

4.2 Major Categories

Category	Mechanism
Hypovolemic	Loss of circulating blood / fluid
Cardiogenic	Failure of cardiac pumping
Distributive	Includes septic and anaphylactic shock
Obstructive	Mechanical obstruction of circulation

4.3 Hypovolemic Shock

Possible causes: major hemorrhage, severe dehydration, burns.

Initial Management

1. Support airway and breathing.
2. Control bleeding.
3. Obtain IV access.

Drug / Fluid	Dose
Normal saline / Ringer lactate	20 mL/kg IV bolus
Tranexamic acid	1 g IV slowly

Blood products may be required in significant hemorrhage.

Vasopressors are advised when blood pressure does not improve appropriately with fluid treatment.

4.4 Cardiogenic Shock

Possible causes: acute myocardial infarction, severe arrhythmia, heart failure. Possible features: severe hypotension, pulmonary edema, raised JVP.

Drug	Dose
Dobutamine	2–20 mcg/kg/min IV infusion
Noradrenaline	0.05–0.3 mcg/kg/min IV
Furosemide (if pulmonary edema present)	20–40 mg IV

⚠ WARNING

- Avoid indiscriminate fluid loading in cardiogenic shock — the failing heart cannot handle an uncontrolled volume load.

4.5 Septic Shock

Possible sources: pneumonia, urinary infection, abdominal infection.

Drug / Fluid	Dose
Crystalloid	30 mL/kg IV within 3 hours
Piperacillin/tazobactam	4.5 g IV every 8 hours
Meropenem	1 g IV every 8 hours
Noradrenaline	0.05–0.3 mcg/kg/min IV

4.6 Anaphylactic Shock

Possible triggers: medicines, foods, insect exposure. Possible findings: breathing difficulty, facial/lip swelling, hypotension.

Drug	Dose
Adrenaline (1:1000)	0.5 mg IM, repeat every 5–15 min
Hydrocortisone	100 mg IV
Chlorpheniramine	10 mg IV

IV normal saline may also be required for circulatory support.

Chapter Summary

Shock is classified as hypovolemic, cardiogenic, septic, anaphylactic, or obstructive, and each type has a different first-line fluid and drug strategy — aggressive fluids for hypovolemic and septic shock, cautious fluids with inotropes for cardiogenic shock, and immediate intramuscular adrenaline for anaphylaxis.

Key Takeaways

- Shock can be present before blood pressure drops — do not wait for hypotension to act.

- Hypovolemic and septic shock both start with fluid resuscitation, but at different rates and volumes.
- Cardiogenic shock needs inotropic support, not large fluid boluses.
- Anaphylaxis is treated with IM adrenaline first, repeated every 5–15 minutes as needed.

✓ **QUICK RECALL**

- Name the four major categories of shock.
- What is the first-line drug, dose, and route for anaphylactic shock?
- Why should fluids be given cautiously in cardiogenic shock?

CHAPTER 5

Respiratory Emergencies



⚠ RED FLAGS

- Severe dyspnea
- Cyanosis
- Markedly abnormal respiratory rate
- Abnormal chest movement
- Reduced oxygen saturation
- Exhaustion
- Altered consciousness

5.1 Acute Asthma

Possible findings: wheeze, chest tightness, breathlessness, difficulty speaking, and a silent chest in severe disease.

Drug	Dose
Salbutamol	2.5–5 mg nebulized, every 20 min × 3 doses
Ipratropium	0.5 mg nebulized
Hydrocortisone	100 mg IV stat
Magnesium sulfate (severe cases)	2 g IV slowly over ~20 min

Oxygen 10–15 L/min by non-rebreather mask is also used.

5.2 Acute COPD Exacerbation

Features may include increased breathlessness, chronic cough, and increased sputum.

 TIP

- Target SpO₂ of 88–92% in COPD — over-oxygenation can suppress the respiratory drive in these patients.

Drug	Dose
Salbutamol	2.5 mg nebulized
Ipratropium	0.5 mg nebulized
Hydrocortisone	100 mg IV
Amoxicillin/clavulanate (if infection suspected)	1.2 g IV every 8 hours

5.3 Pulmonary Edema

Features may include severe breathlessness, bilateral crepitations, and pink or frothy sputum.

Drug	Dose
Furosemide	40 mg IV slowly
Nitroglycerin	5 mcg/min IV infusion (monitor BP)
Morphine (for anxiety)	2–4 mg IV

Oxygen or CPAP support is used alongside these medications.

5.4 Tension Pneumothorax

Possible signs: severe respiratory distress, hypotension, tracheal deviation.

★ KEY POINT

- Emergency needle decompression with a 14G cannula at the 2nd intercostal space, followed by chest-tube insertion.

5.5 Aspiration

Initial priorities: airway protection, suction, oxygenation.

Drug	Dose
Piperacillin/tazobactam (aspiration pneumonia)	4.5 g IV every 8 hours

5.6 Respiratory Failure

Defined as PaO_2 below 60 mmHg and/or PaCO_2 above 50 mmHg, and distinguished as hypoxic or hypercapnic. Management may progress from oxygen/NIV to intubation and ventilation in severe disease.

Chapter Summary

Respiratory emergencies share a common early pattern — bronchodilators and steroids for airway disease, diuretics and nitrates for pulmonary edema, and immediate decompression for tension pneumothorax — with oxygen targets and thresholds that differ by diagnosis.

Key Takeaways

- Asthma and COPD both receive nebulized salbutamol and ipratropium, but COPD needs a lower SpO_2 target.

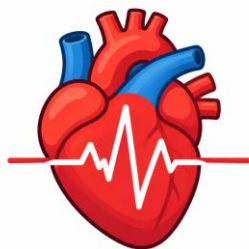
- Severe asthma adds IV magnesium sulfate when initial treatment is insufficient.
- Tension pneumothorax is a clinical diagnosis treated immediately with needle decompression — do not wait for imaging.
- Respiratory failure is defined by specific PaO_2 and PaCO_2 thresholds, not by symptoms alone.

✓ QUICK RECALL

- What SpO_2 range is targeted in COPD and why?
- Where is needle decompression performed for tension pneumothorax?
- What are the PaO_2 and PaCO_2 thresholds that define respiratory failure?

CHAPTER 6

Cardiac Emergencies



6.1 Acute Coronary Syndrome

Possible symptoms: central or left-sided pressure or pain, radiation to arm, jaw, or back, dyspnea, sweating, nausea/vomiting. Obtain an ECG rapidly.

Drug	Dose
Aspirin	300 mg chewed
Nitroglycerin (if BP > 90 mmHg)	0.5 mg sublingual
Morphine	2–4 mg IV slowly
Clopidogrel	300 mg PO
Heparin	5000 IU IV bolus
Atorvastatin	80 mg stat

Oxygen is added when SpO₂ is below 94%. Primary PCI is the preferred reperfusion strategy; weight-based tenecteplase is used for thrombolysis where PCI is unavailable.

6.2 Tachyarrhythmias

Supraventricular Tachycardia (SVT)

Start with appropriate vagal manoeuvres when indicated.

Drug	Dose
Adenosine (1st dose)	6 mg rapid IV
Adenosine (if unsuccessful)	12 mg rapid IV

Ventricular Tachycardia

Drug	Dose
Amiodarone	300 mg IV bolus

Defibrillation is used for unstable VT.

6.3 Bradyarrhythmia

Drug	Dose
Atropine	0.5 mg IV every 3–5 min, maximum 3 mg

6.4 Acute Heart Failure

Drug	Dose
Furosemide	40 mg IV
Nitroglycerin	5 mcg/min IV infusion
Dobutamine (if low BP present)	2–20 mcg/kg/min IV

6.5 Hypertensive Emergency

Characterized by severe hypertension with organ damage.

Drug	Dose
Labetalol	20 mg IV bolus
Nitroglycerin	5 mcg/min IV infusion
Sodium nitroprusside	ICU setting

6.6 Cardiac Tamponade

Remember Beck's triad:

- Hypotension
- Raised JVP
- Muffled heart sounds

Definitive emergency management may require pericardiocentesis.

Chapter Summary

Cardiac emergencies are separated by rhythm and mechanism: ACS needs rapid antiplatelet and reperfusion therapy, tachyarrhythmias are treated according to rate and stability, bradycardia responds to atropine, and tamponade is recognized by Beck's triad and relieved by pericardiocentesis.

Key Takeaways

- Aspirin 300 mg chewed is a near-immediate step in suspected ACS.
- Adenosine for SVT is given as a rapid IV push — 6 mg first, then 12 mg if needed.
- Atropine for bradycardia is capped at a 3 mg maximum.
- Beck's triad — hypotension, raised JVP, muffled heart sounds — points to cardiac tamponade.

✓ QUICK RECALL

- What is the first-line antiplatelet dose in suspected ACS?
- What are the two adenosine doses used for SVT, in order?
- What are the three components of Beck's triad?

CHAPTER 7

Neurological Emergencies



⚠ RED FLAGS

- Sudden unconsciousness
- Seizures
- New unilateral weakness
- Speech difficulty
- Sudden severe headache
- Rapidly falling GCS

Always consider blood glucose early in any neurological emergency.

7.1 Seizures

Immediate care: protect from injury, position appropriately, support airway, give oxygen when indicated, and check glucose.

Drug	Dose
Diazepam	10 mg IV slowly
Lorazepam (alternative)	4 mg IV slowly
Phenytoin (if seizure continues)	15–20 mg/kg IV slowly

7.2 Status Epilepticus

Defined using a seizure duration greater than 5 minutes.

Lorazepam 4 mg IV



Phenytoin 20 mg/kg IV



Midazolam infusion if refractory

Refractory cases may require midazolam infusion and advanced airway/ventilatory management.

7.3 Stroke

Use FAST:

- F — Face drooping
- A — Arm weakness
- S — Speech difficulty
- T — Time to hospital

Differentiate ischemic from hemorrhagic stroke with appropriate brain imaging.

Ischemic Stroke

Drug	Dose
Aspirin (after CT)	150–300 mg PO/NG
Atorvastatin	40–80 mg
Alteplase (eligible patients, within 4.5 hours)	0.9 mg/kg IV

Hemorrhagic Stroke

Drug	Dose
Labetalol (BP control)	20 mg IV

Neurosurgical consultation is required.

7.4 Altered Sensorium / Coma

Possible causes: hypoglycemia, stroke, poisoning, infection.

★ KEY POINT

- For unknown cause, the “emergency cocktail” is: Dextrose 25% 100 mL IV, Thiamine 100 mg IV, Naloxone 0.4–2 mg IV.

7.5 Raised Intracranial Pressure

Possible findings: headache, vomiting, papilledema, reduced GCS.

Drug / Measure	Dose
Head elevation	~30°
Mannitol	0.5–1 g/kg IV
Hypertonic saline (ICU use)	ICU setting

7.6 Meningitis / Encephalitis

Possible presentation: fever, neck stiffness, altered consciousness.

Drug	Dose
Ceftriaxone	2 g IV every 12 hours
Vancomycin	Weight-based
Dexamethasone	10 mg IV

Chapter Summary

Neurological emergencies share two early habits regardless of diagnosis: check blood glucose immediately, and use FAST and GCS to decide how fast imaging and definitive treatment are needed.

Seizure control, stroke thrombolysis windows, and empirical meningitis antibiotics each carry a specific time-sensitive threshold.

Key Takeaways

- Status epilepticus is defined by a seizure lasting more than 5 minutes.
- Alteplase for ischemic stroke must be given within 4.5 hours, after imaging rules out hemorrhage.
- The “emergency cocktail” for unexplained coma covers glucose, thiamine, and opioid reversal together.
- Raised ICP management starts with head elevation before escalating to mannitol.

✓ QUICK RECALL

- What seizure duration defines status epilepticus?
- What three drugs make up the “emergency cocktail” for coma of unknown cause?
- What is the treatment time window for alteplase in eligible ischemic stroke?

CHAPTER 8

Poisoning & Overdose



The priorities in any poisoning case follow a fixed sequence:

ABCDE



Identify poison



Limit further exposure/absorption when appropriate



Antidote when indicated



Supportive care

8.1 Organophosphate Poisoning

Remember SLUDGE:

- Salivation
- Lacrimation
- Urination
- Diarrhea
- Gastrointestinal cramps
- Emesis

Drug	Dose
Atropine	2–5 mg IV, repeat every 5–10 min until chest is dry
Pralidoxime	1–2 g IV over 30 min, then 1 g IV every 8–12 hours

8.2 Drug Overdose

Situation	Drug	Dose
General decontamination (within 1 hour)	Activated charcoal	1 g/kg orally
Opioid overdose	Naloxone	0.4–2 mg IV/IM
Benzodiazepine overdose	Flumazenil	0.2 mg IV slowly
Paracetamol overdose	N-acetylcysteine	IV protocol according to weight

8.3 Alcohol Poisoning

Possible findings: slurred speech, hypoglycemia, respiratory depression.

Drug	Dose
Dextrose 25%	100 mL IV
Thiamine	100 mg IV

IV fluids and airway protection are also provided.

8.4 Corrosive Poisoning

⚠ WARNING

- Do not perform gastric lavage in corrosive poisoning.

Initial approach: mouth rinse, IV fluids, appropriate specialist assessment.

Drug	Dose
Pantoprazole	40 mg IV

8.5 Snakebite

Possible manifestations: local swelling, bleeding, neuromuscular paralysis.

Drug	Dose
Anti-snake venom (initial dose)	10 vials IV
Adrenaline (if anaphylaxis occurs)	0.5 mg IM

Chapter Summary

Poisoning management always begins with ABCDE and identification of the agent, then branches by toxin: atropine and pralidoxime for organophosphates, specific antidotes for opioid/benzodiazepine/paracetamol overdose, and antivenom for snakebite — with gastric lavage specifically avoided in corrosive ingestion.

Key Takeaways

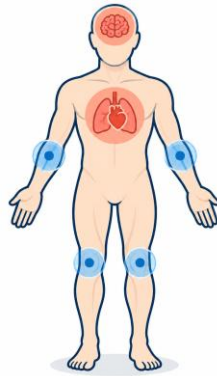
- SLUDGE symptoms point to organophosphate poisoning; atropine is titrated to a dry chest, not a fixed dose.
- Activated charcoal is most useful within one hour of ingestion.
- Naloxone, flumazenil, and N-acetylcysteine are antidotes matched to opioids, benzodiazepines, and paracetamol respectively.
- Gastric lavage is contraindicated in corrosive poisoning.

✓ **QUICK RECALL**

- What does the SLUDGE mnemonic stand for?
- What is the antidote and dose for opioid overdose?
- Why is gastric lavage avoided in corrosive ingestion?

CHAPTER 9

Trauma Management



Trauma management begins with the same structure as the general primary survey, adapted for injury:

A — Airway + cervical-spine protection



B — Breathing



C — Circulation + hemorrhage control



D — Disability / GCS



E — Exposure

9.1 Head Injury

Look for reduced consciousness, vomiting, seizures, falling GCS, and focal neurological signs.

Drug	Dose
Mannitol	0.5–1 g/kg IV slowly
Phenytoin	15 mg/kg IV

Drug	Dose
Paracetamol	1 g IV

TIP

- Urgent CT is indicated with GCS below 13, repeated vomiting, or any focal neurological deficit.

9.2 Chest Trauma

Consider rib fractures, pneumothorax, and hemothorax.

Drug	Dose
Tramadol	50–100 mg IV
Paracetamol	1 g IV

For tension pneumothorax: 14G needle decompression at the 2nd intercostal space, followed by chest-tube insertion.

9.3 Abdominal Trauma

Possible findings: abdominal pain, guarding/rigidity, hypotension.

Management: keep the patient NPO, establish IV access/fluids, perform a FAST ultrasound, and obtain surgical assessment where indicated.

Drug / Fluid	Dose
Normal saline / Ringer lactate	20 mL/kg IV
Ceftriaxone (prophylactic)	1–2 g IV

9.4 Musculoskeletal Trauma

Look for pain, swelling, deformity, and neurovascular compromise. Immobilize or splint appropriately.

Drug	Dose
Diclofenac	75 mg IM
Tramadol	50 mg IV

9.5 Polytrauma / Major Hemorrhage

Priorities: airway/oxygenation, hemorrhage control, circulatory resuscitation, blood products where required, and rapid definitive trauma care.

Drug / Fluid	Dose
Crystalloids	1–2 L IV
Tranexamic acid	1 g IV within 3 hours

Blood transfusion is given when required.

Chapter Summary

Trauma follows the familiar ABCDE order with two injury-specific additions: cervical-spine protection during airway management, and active hemorrhage control during circulation. Each body region then has its own red-flag findings and first-line analgesia or intervention.

Key Takeaways

- Cervical-spine protection is part of the airway step in every trauma patient.
- GCS below 13, repeated vomiting, or a focal deficit after head injury are urgent CT indications.
- Tension pneumothorax is decompressed immediately with a 14G needle at the 2nd intercostal space.
- Tranexamic acid should be given within 3 hours in major hemorrhage.

✓ **QUICK RECALL**

- What is added to the standard ABCDE sequence for a trauma patient?
- What GCS value prompts an urgent CT after head injury?
- What is the time window for giving tranexamic acid in major hemorrhage?

CHAPTER 10

Pediatric Emergencies



Children may deteriorate rapidly. Key principles:

- Medication is commonly weight-based
- Hypoglycemia can occur rapidly
- Respiratory compromise is important to identify early
- Fluid therapy requires pediatric calculations

10.1 Pediatric Assessment Triangle

Evaluate three domains at a glance:

- Appearance
- Breathing
- Circulation

10.2 Pediatric Respiratory Distress

Possible causes: bronchiolitis, pneumonia, asthma, foreign body.

Drug	Dose
Salbutamol	0.15 mg/kg nebulized, maximum 5 mg
Hydrocortisone	5 mg/kg IV

Oxygen is given at 10–15 L/min.

10.3 Febrile Seizure

Reference age range: 6 months to 5 years.

Drug	Dose
Diazepam	0.3 mg/kg IV or 0.5 mg/kg PR
Paracetamol (fever control)	10–15 mg/kg

10.4 Pediatric Dehydration / Shock

Signs can include sunken eyes, dry tongue, reduced urine output, and poor perfusion.

Drug / Fluid	Dose
Normal saline / Ringer lactate	20 mL/kg IV bolus, repeated as clinically required
Dopamine (if shock persists)	5–10 mcg/kg/min IV infusion

10.5 Pediatric Sepsis

Drug / Fluid	Dose
Fluid bolus	20 mL/kg IV
Ceftriaxone	50–75 mg/kg IV once daily
Amikacin	15 mg/kg IV

10.6 Pediatric Hypoglycemia

Drug	Dose
Dextrose 10%	5 mL/kg IV bolus

10.7 Pediatric Poisoning

Drug	Dose
Activated charcoal	1 g/kg orally
Naloxone (opioid poisoning)	0.1 mg/kg IV

Chapter Summary

Pediatric emergencies are managed with the same recognition principles as adult care, but every drug and fluid dose is weight-based, and hypoglycemia and respiratory compromise are watched for more aggressively because children deteriorate faster.

Key Takeaways

- The Pediatric Assessment Triangle — appearance, breathing, circulation — gives a rapid first impression.
- Febrile seizures are defined within a specific age range of 6 months to 5 years.
- Fluid boluses in pediatric shock and sepsis both use 20 mL/kg IV as the starting dose.
- Never substitute an adult drug dose for a pediatric calculation — doses are weight-based throughout.

✓ QUICK RECALL

- What three domains make up the Pediatric Assessment Triangle?
- What is the standard pediatric fluid bolus dose in shock?
- What age range defines a febrile seizure in this reference?

CHAPTER 11

Obstetric Emergencies



Major emergencies covered in this chapter include:

- Postpartum hemorrhage
- Eclampsia / severe preeclampsia
- Antepartum hemorrhage
- Ruptured ectopic pregnancy
- Obstructed labor
- Sepsis in pregnancy

11.1 Postpartum Hemorrhage (PPH)

Defined as blood loss greater than 500 mL after vaginal delivery, or greater than 1000 mL after cesarean delivery.

Remember the 4 Ts:

- Tone — uterine atony
- Trauma — tears
- Tissue — retained placental tissue
- Thrombin — coagulation problem

Initial Measures

1. Call for help.
2. Uterine massage.
3. Two large-bore IV lines.
4. Assess and resuscitate circulation.

Drug	Dose
Oxytocin	10 IU IM or 20 IU IV infusion
Methylergometrine	0.2 mg IM (check BP first)
Misoprostol	800–1000 mcg PR
Tranexamic acid	1 g IV slowly

11.2 Eclampsia

Possible findings: severe hypertension, headache, seizures. Severe BP threshold: $\geq 160/110$ mmHg.

Magnesium Sulfate — Pritchard Regimen

Phase	Dose
Loading	4 g IV + 10 g IM
Maintenance	5 g IM every 4 hours

BP Control

Drug	Dose
Labetalol	20 mg IV bolus, repeated as required

11.3 Antepartum Hemorrhage

Possible causes: placenta previa, placental abruption.

⚠ WARNING

- Do not perform a PV (per vaginal) examination in suspected antepartum hemorrhage.

Provide appropriate resuscitation, blood products where required, and urgent obstetric assessment.

11.4 Ruptured Ectopic Pregnancy

Possible findings: sudden severe abdominal pain, hypotension, shoulder-tip pain.

Priorities: rapid IV access, fluid/blood resuscitation, emergency surgical management.

11.5 Obstructed Labor

Possible findings: prolonged labor, maternal exhaustion, fetal distress.

Drug	Dose
Ceftriaxone	1 g IV
Metronidazole	500 mg IV

Definitive obstetric management follows.

11.6 Sepsis in Pregnancy

Possible features: fever, tachycardia, foul-smelling discharge.

Drug / Fluid	Dose
Crystalloid	30 mL/kg IV
Piperacillin/tazobactam	4.5 g IV every 8 hours
Clindamycin	900 mg IV

Chapter Summary

Obstetric emergencies combine standard resuscitation with pregnancy-specific thresholds and drugs: the 4 Ts guide PPH management, the Pritchard magnesium regimen controls eclamptic seizures, and antepartum hemorrhage specifically avoids PV examination until placenta previa is excluded.

Key Takeaways

- PPH is defined by a specific blood-loss threshold that differs for vaginal versus cesarean delivery.
- The 4 Ts — Tone, Trauma, Tissue, Thrombin — cover the causes of PPH.
- The Pritchard magnesium sulfate regimen has a distinct loading and maintenance phase.
- PV examination is withheld in suspected antepartum hemorrhage until placenta previa is excluded.

✓ QUICK RECALL

- What blood-loss volumes define PPH after vaginal versus cesarean delivery?
- What are the loading and maintenance doses in the Pritchard magnesium regimen?
- Why is a PV examination avoided in suspected antepartum hemorrhage?

CHAPTER 12

Emergency Drug Quick Reference



This chapter consolidates the drug doses already introduced in earlier chapters into one indication-based quick-reference section, organized by clinical category so a dose can be found quickly without re-reading an entire chapter.

12.1 Resuscitation

Drug	Indication	Dose
Adrenaline	Cardiac arrest	1 mg IV every 3–5 min
Adrenaline	Anaphylaxis	0.5 mg IM (1:1000)
Atropine	Bradycardia	0.5 mg IV, max 3 mg
Atropine	OP poisoning	2–5 mg IV, titrated to response
Noradrenaline	Septic shock / hypotension	0.05–0.3 mcg/kg/min IV

12.2 Respiratory

Drug	Dose
Salbutamol	2.5–5 mg nebulized
Ipratropium	0.5 mg nebulized
Hydrocortisone	100 mg IV

12.3 Cardiac

Drug	Dose
Aspirin	300 mg chewed
Nitroglycerin	0.5 mg SL, or 5 mcg/min IV infusion
Amiodarone	300 mg IV bolus

12.4 Neurological

Drug	Dose
Diazepam	10 mg IV slowly
Lorazepam	4 mg IV
Phenytoin	15–20 mg/kg IV slowly

12.5 Shock / Trauma

Drug	Dose
Tranexamic acid	1 g IV slowly
Dopamine	5–10 mcg/kg/min IV
Dobutamine	2–20 mcg/kg/min IV

12.6 Antidotes

Drug	Dose
Pralidoxime	1–2 g IV

Drug	Dose
Naloxone	0.4–2 mg IV/IM
Activated charcoal	1 g/kg orally

12.7 Obstetric

Drug	Dose
Oxytocin	10 IU IM / 20 IU IV infusion
Magnesium sulfate	Pritchard regimen (see Chapter 11)
Misoprostol	800–1000 mcg PR

12.8 Metabolic

Drug	Dose
Dextrose 25% (adult)	100 mL IV
Dextrose 10% (child)	5 mL/kg IV
Regular insulin (DKA infusion)	0.1 U/kg/hour IV

12.9 Analgesia

Drug	Dose
Morphine	2–4 mg IV
Tramadol	50–100 mg IV

TIP

- Use this chapter as a fast lookup during real practice, but always cross-check the full chapter discussion for context, contraindications, and monitoring before administering any drug.

Chapter Summary

This quick-reference chapter regroups every drug dose already covered in Chapters 2 through 11 by clinical category — resuscitation, respiratory, cardiac, neurological, shock/trauma, antidotes, obstetric, metabolic, and analgesia — for rapid lookup.

Key Takeaways

- This chapter is a lookup tool, not a substitute for the clinical context in the earlier chapters.
- Doses are grouped by category rather than by chapter, so related drugs sit next to each other.
- Always confirm a dose here against the fuller discussion before administering it.

✓ QUICK RECALL

- What is the adrenaline dose for cardiac arrest versus anaphylaxis?
- What is the maximum cumulative atropine dose in bradycardia?
- What insulin infusion rate is used in DKA?

CHAPTER 13

Emergency Investigations



Investigations should help answer urgent clinical questions without unnecessarily delaying stabilization. Emergency investigations are grouped into bedside, laboratory, imaging, and special tests.

13.1 Bedside Tests

Pulse Oximetry

Measures oxygen saturation. Normal range: 94–100%.

Random Blood Sugar

Normal range: 70–140 mg/dL. Below 70 mg/dL is categorized as hypoglycemia. Particularly important in altered consciousness and seizures.

ECG

Useful in chest pain, palpitations, and syncope.

★ KEY POINT

- Chest pain → ECG within 10 minutes.

13.2 Laboratory Tests

Test	What It Shows
CBC	Hemoglobin (anemia/blood loss), TLC (infection), platelets (bleeding risk)
Electrolytes (Na ⁺ , K ⁺)	Important in arrhythmias, seizures, metabolic disturbances
Renal function (urea/creatinine)	Relevant in shock, poisoning, renal dysfunction
Liver function	Relevant to liver disease and selected toxicological presentations
ABG	Provides pH, PaO ₂ , PaCO ₂ — useful in respiratory failure and metabolic acidosis

13.3 Imaging

Imaging	Use
Chest X-ray	Pneumonia, pneumothorax, pulmonary edema
CT brain	Selected stroke and head injury presentations
CT abdomen	Selected trauma / acute abdominal presentations
FAST ultrasound	Rapid bedside assessment for internal bleeding in trauma

13.4 Cardiac Investigations

Test	Use
Troponin I/T	Acute myocardial injury evaluation
Echocardiography	Heart failure, tamponade, selected cardiovascular emergencies

13.5 Infection Workup

Blood culture should be obtained before antibiotics where feasible in sepsis. CSF analysis is used in meningitis evaluation — brain CT is obtained first where raised intracranial pressure is suspected.

13.6 Time-Critical Quick View

Presentation	First Investigation
Chest pain	ECG
Stroke	CT brain
Seizure / altered consciousness	Blood glucose
Trauma	FAST where indicated
Shock	ABG + lactate

Chapter Summary

Emergency investigations are chosen to answer one urgent question at a time — bedside tests first, laboratory and imaging next — and several presentations have a fixed first investigation that should never be delayed, such as an ECG within 10 minutes for chest pain.

Key Takeaways

- Pulse oximetry, blood glucose, and ECG are rapid bedside tests used before any lab result returns.
- Blood cultures should be drawn before antibiotics in sepsis whenever feasible.
- Chest pain, stroke, seizure, trauma, and shock each map to one specific first investigation.

✓ **QUICK RECALL**

- What blood glucose value defines hypoglycemia in this reference?
- Within what time frame should an ECG be obtained for chest pain?
- What is the first investigation of choice in suspected stroke?

CHAPTER 14

Triage



Triage assigns priority according to the urgency and severity of a patient's condition. The purpose is not simply to treat people in arrival order — it is to ensure that patients with the greatest immediate threat receive appropriate attention first.

14.1 The Four Triage Categories

Category	Meaning	Examples / Action
RED — Immediate	Greatest immediate threat	Cardiac arrest, severe trauma, shock, unconsciousness, severe respiratory distress — immediate resuscitation
YELLOW — Urgent	Serious but currently more stable	Moderate trauma, stable chest pain, fractures — treatment within ~30–60 minutes
GREEN — Non-urgent	Stable, no immediate threat	Minor injury, mild fever, simple pain
BLACK — Dead / expectant	Used in mass-casualty settings	Deceased patients or those categorized as expectant under the applicable disaster-triage system

14.2 START Triage

START stands for Simple Triage And Rapid Treatment. Key criteria:

- Walking patient → GREEN
- Respiratory rate above 30 → RED
- No radial pulse → RED
- Unable to follow commands → RED

14.3 Pediatric Triage

JumpSTART is the pediatric adaptation of START, incorporating breathing, pulse, and mental status specifically calibrated for children.

14.4 Re-Triage

Triage is dynamic — a patient initially categorized as stable can deteriorate. Reassess regularly.

Chapter Summary

Triage sorts patients into four color-coded categories — Red, Yellow, Green, Black — using simple, fast criteria like respiratory rate, pulse, and the ability to walk or follow commands, and the category assigned at arrival is reassessed continuously rather than fixed.

Key Takeaways

- RED means immediate resuscitation; YELLOW allows treatment within 30–60 minutes.
- A respiratory rate above 30 or an absent radial pulse both trigger a RED category under START.
- JumpSTART adapts the same logic for pediatric patients.
- Triage categories must be reassessed — a stable patient can deteriorate at any time.

✓ **QUICK RECALL**

- What do the four triage colors represent?
- What respiratory rate threshold triggers a RED category in START?
- What is the pediatric version of START called?

CHAPTER 15

Medico-Legal Essentials



A medico-legal case is one in which illness or injury may involve legal investigation or police/court processes. Common examples include:

- Road-traffic accidents
- Assault
- Poisoning
- Burns
- Sexual assault
- Hanging/drowning
- Unknown unconscious patients

15.1 Documentation

Record the following for every medico-legal case:

- Date/time
- Patient details
- History as reported
- Examination findings
- Treatment administered
- Drug/dose/time

- Clinical response
- Referral/discharge condition

⚠ WARNING

- Avoid unclear alteration of records. Sign appropriately according to institutional requirements.

15.2 Consent

Three forms of consent are distinguished:

- Implied consent
- Informed consent
- Written consent

Implied consent typically applies to an unconscious patient requiring life-saving care.

✍ TIP

- Consent law and medico-legal duties vary by jurisdiction and circumstance — always check this section against the law and hospital policy applicable where you practice.

15.3 Police Intimation

Situations that typically require written police intimation include accidents, poisoning, burns, assault, and suspected homicide/suicide. Appropriate written intimation and hospital documentation should follow.

15.4 Evidence Preservation

Depending on the case, relevant material may include blood-stained clothing, gastric samples, vomitus, and foreign bodies. Proper identification, labeling, documentation, and chain-of-custody procedures are important.

Chapter Summary

Medico-legal cases require the same clinical urgency as any emergency, plus four additional disciplines: precise documentation, correct identification of the applicable form of consent, timely police intimation where required, and careful preservation of physical evidence with a clear chain of custody.

Key Takeaways

- Document every detail — time, findings, treatment, drug/dose/time, and outcome — without altering the record.
- Implied consent applies to unconscious patients needing life-saving treatment; other situations need informed or written consent.
- Police intimation applies to accidents, assault, poisoning, burns, and suspected homicide/suicide.
- Evidence such as clothing or samples must be labeled and tracked with a clear chain of custody.

✓ QUICK RECALL

- What type of consent applies to an unconscious patient needing life-saving care?
- Name three situations that require police intimation.
- Why is chain-of-custody documentation important for preserved evidence?

CHAPTER 16

Emergency Medicine Career Pathways



Emergency medicine requires rapid decision-making, teamwork, resuscitation skills, the ability to function in high-pressure environments, and continuous clinical learning. Opportunities exist for doctors, nurses, and paramedical/EMT professionals.

16.1 After MBBS

Emergency Medical Officer

Possible workplaces: private hospitals, trauma centers, corporate hospitals. Responsibilities may include initial assessment, resuscitation, and stabilization.

Junior Resident — Emergency Department

Teaching hospitals and medical colleges can provide extensive emergency exposure.

ER Doctor / Hospitalist

May provide shift-based emergency coverage.

16.2 Postgraduate Training

Pathway	Notes
MD Emergency Medicine	Postgraduate degree route
DNB Emergency Medicine	Postgraduate diploma route
Diploma / Fellowship pathways	Various shorter postgraduate options

 TIP

- Skills-based training such as ACLS, ATLS, and PALS is highlighted as valuable alongside formal postgraduate pathways.

Chapter Summary

Emergency medicine offers entry points for doctors, nurses, and paramedical staff, ranging from emergency medical officer roles straight after MBBS to postgraduate pathways such as MD or DNB Emergency Medicine, supported by skills courses like ACLS, ATLS, and PALS.

Key Takeaways

- Entry-level emergency roles are open immediately after MBBS, before postgraduate specialization.
- MD and DNB Emergency Medicine are the main postgraduate degree routes.
- ACLS, ATLS, and PALS are skills-based courses that complement formal postgraduate training.
- Emergency medicine careers exist for doctors, nurses, and paramedical/EMT staff alike.

✓ **QUICK RECALL**

- What are two postgraduate degree pathways in emergency medicine?
- Name three skills-based courses relevant to emergency medicine training.
- What professional groups, besides doctors, have career pathways in emergency medicine?

FINAL SECTION

Rapid Emergency Reference

This section condenses the entire guide into fast, scannable cards for the moments when there is no time to flip back through chapters.

The First-Minutes Checklist

A — Airway

- Patent?
- Protected?
- Obstruction?
- Trauma / C-spine concern?

B — Breathing

- Rate?
- Effort?
- SpO₂?
- Chest movement?
- Air entry?

C — Circulation

- Pulse?
- BP?
- Perfusion?
- Bleeding?
- IV access required?

D — Disability

- GCS?
- Pupils?

- Blood glucose?
- Seizure?
- Focal neurological deficit?

E — Exposure

- Bleeding?
- Burns?
- Fractures?
- Rash?
- Hidden injury?
- Temperature?

Cardiac Arrest Quick Card

Recognize arrest



Call for help / activate response



CPR 100–120/min



Compression depth 5–6 cm



Attach defibrillator



Identify shockable / non-shockable rhythm



Treat according to arrest pathway



Search for reversible causes



Continue until ROSC / appropriate termination decision

Shock — Quick Differentiation

Type	Think
Hypovolemic	Volume / blood loss
Cardiogenic	Pump failure
Septic	Severe infection + circulatory dysfunction
Anaphylactic	Allergic reaction + airway/breathing/circulation involvement
Obstructive	Mechanical obstruction to circulation

Stroke Quick Card

- F — Face
- A — Arm
- S — Speech
- T — Time

ABCDE



Glucose



Establish onset / last-known-well



Urgent brain imaging



Stroke pathway

Seizure Quick Card

- Protect patient from injury.
- Do not place inappropriate objects in the mouth.
- Support airway / breathing.
- Check glucose.
- Time the seizure.
- Use appropriate medication when indicated.

- Consider status epilepticus when seizure is prolonged according to the applicable protocol.

Poisoning Quick Card

ABCDE



Identify substance if possible



Route + amount + time



Prevent further exposure



Appropriate decontamination only when indicated



Antidote when appropriate



Supportive care



Monitor + reassess

Trauma Quick Card

Airway + C-spine



Breathing



Circulation + hemorrhage control



Disability / GCS



Exposure

Then proceed to secondary assessment and appropriate imaging/definitive management.

Pediatric Quick Card

★ KEY POINT

- Weight matters. Drug and fluid doses are commonly weight-based. Never automatically substitute an adult dose for a pediatric calculation.

Check: appearance, breathing, circulation, and glucose where indicated.

Obstetric Red Flags

Urgent assessment for:

- Heavy vaginal bleeding
- Seizure in pregnancy
- Severe hypertension with concerning symptoms
- Shock
- Severe abdominal pain in pregnancy
- Fetal distress
- Suspected sepsis
- Prolonged / obstructed labor

Emergency Drug Snapshot

Drug	Indication	Dose
Adrenaline	Cardiac arrest	1 mg IV
Adrenaline	Anaphylaxis	0.5 mg IM
Aspirin	ACS	300 mg
Atropine	Bradycardia	0.5 mg IV
Amiodarone	VF/VT	300 mg IV
Diazepam	Seizure	10 mg IV
Lorazepam	Status epilepticus	4 mg IV

Drug	Indication	Dose
Naloxone	Opioid overdose	0.4–2 mg IV/IM
Tranexamic acid	Major bleeding	1 g IV
Oxytocin	PPH	10 IU IM
Salbutamol	Asthma/COPD	2.5–5 mg nebulized
Hydrocortisone	Multiple emergency uses	100 mg IV
Mannitol	Raised ICP	0.5–1 g/kg IV

Final Takeaway

Emergency care is not simply about memorizing drug names. A reliable emergency approach combines rapid recognition, structured ABCDE assessment, immediate stabilization, correct medication and dose, focused investigation, and continuous reassessment. The patient's condition can change rapidly, so every intervention should be followed by reassessment.

★ KEY POINT

- The emergency-care cycle: ASSESS → ACT → MONITOR → REASSESS.